THERMAL MANAGEMENT

Closed-Loop Digester Mesophilic Stability Circuit

BEIJING PRESTIGE AUTOMATION PST-800

THE MESOPHILIC MANDATE

Maintaining a rigid environment for bacterial colony survival against Highveld winter threats.

HIGHVELD WINTER THREATS

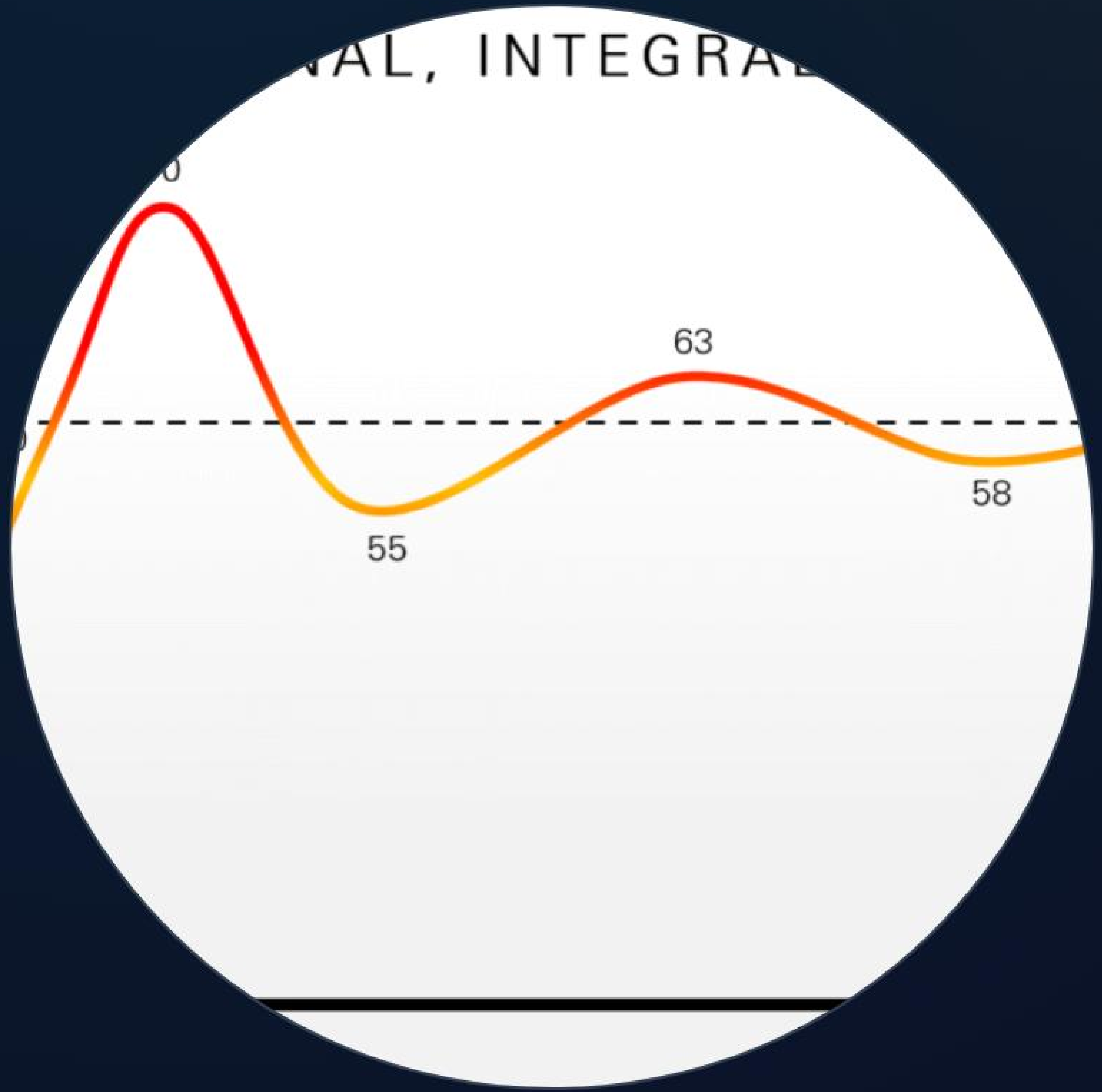
Zero-Watt Thermal Loss

Highveld winters drop temperatures to sub-zero. Without active thermal management, the digester core freezes, killing the methanogenic bacteria and halting energy production for months.

Mesophilic Zone

Bacteria require a constant **35°C to 38°C**. Even a 2°C drop significantly reduces biogas yield, while fluctuations above 42°C result in rapid bacterial mortality.

THE CONTROLLER INTERFACE



PST-800 Determinant Control

The Beijing Prestige PST-800 acts as the sovereign nerve, processing 4-20mA signals from the probe array. It executes a high-speed PID algorithm to modulate heat injection with 0.1°C precision.

SENSORS: PT100 PROBE ARRAY

High-Accuracy Thermometry

Embedded PT100 thermocouple arrays measure core temperature at three distinct depths. This ensures the PLC detects thermal stratification before it affects the entire tank.

Data is filtered via shielded cabling to prevent EMI interference from the co-generation engine room.

LOGIC: THE PID ALGORITHM

P.I.D.

PROPORTIONAL • INTEGRAL • DERIVATIVE

Zero-Drift Execution

Instead of simple ON/OFF switching, the PLC calculates the **rate of change**. It predicts heat loss based on ambient temperature drops and adjusts the valve position before the core temperature even begins to fall.

THE MIXING VALVE

A motorized three-way mixing valve is the heart of the circuit. It blends high-temperature steam/fluid from the co-generation loop with cool return fluid from the digester floor grids.

This allows the system to maintain a steady heat flux without thermal shock to the tank floor.

RECOVERY: CO-GENERATION LOOP

Zero-Cost Thermal Energy

Heat is sourced from the exhaust-gas-to-steam heat exchanger on the co-generation engine. By recycling waste heat, we maintain the 38°C mesophilic zone with zero additional electrical draw from the microgrid storage.

SAFETY: BYPASS INTERLOCKS

THRESHOLD	LOGIC TRIGGER	AUTOMATED ACTION
< 34°C	Thermal Deficit	Open Mixing Valve to 100% Flow
> 40°C	Thermal Surge	Close Heat Intake; Open External Radiator
> 42°C	Critical Alarm	Total Bypass Lockout; Audible Warning (Q4)
Sensor Fail	Signal Loss	Enter Default 50% Safe-Mix Position

STRUCTURAL ASSISTANCE

- **50mm PIR Insulation:** Rigid panels surrounding the digester shell to minimize radiant heat loss to the Highveld air.
- **Wind Scrim:** Outer cladding designed to prevent convective cooling from winter drafts.
- **Solar Gain:** Passive thermal mass tracking to assist the PID loop during daylight hours.

CALIBRATION & HEALTH

Probe Validation

Sensor drift can cause hidden cooling. Calibrate PT100 probes quarterly against a verified digital reference.

Valve Seating

Inspect mixing valve gaskets for calcium buildup. A stuck valve in winter equals a total system death.

System Locked

Thermal Equilibrium Verified — 北京尊贵自动化 PST-800

LWANDILE ENGINEERING PROJECTS

IMAGE SOURCES

https://media.maplesystems.com/wp-content/uploads/2024/05/3056-PID-1-Temp-Graph-1024x617.png

Source: maplesystems.com

https://www.worldindustrialstore.com/sites/default/files/PT100.jpg

Source: www.worldindustrialstore.com

https://rapidcontrols.in/wp-content/uploads/2025/07/3-way-Motorized-Globe-Valve-Mixing-by-rapidcontrols.in_.jpg

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